

Practice for Introduction to JavaScript: DOM

Week 03.

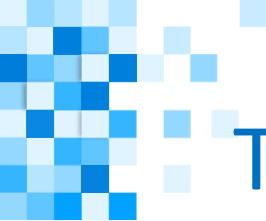


SOFTWARE CLUSTER



Objectives

- Describe the basic structure of the Document Object Model (DOM);
- Describe the way HTML elements can be accessed from the DOM using JavaScript;
- Demonstrate how to write scripts in web page using JavaScript to create and control behaviours of the web page.



Tools Required

- Visual Studio Code
- Node.js
- Browser
 - Google Chrome (recommended)
 - Mozilla Firefox



USING THE GOOGLE MAPS API



What is an API?

- Application Programming Interface (API)
- A way to use someone's code in your code.
- They are usually written as functions or methods.
- They provide certain functionalities.
- Many services offer JavaScript APIs:
 - Google Maps
 - Bing Maps
 - YouTube
 - Vimeo
 - SoundCloud
- To look for more, search for "<service> JavaScript API" in your browser.

Obtaining API Key

- Some APIs are free but some are to be paid for
- To use someone's APIs, you need to obtain a key.
- For Google Maps API, go to the following URL:
 - <https://developers.google.com/maps/documentation/javascript/get-api-key>
 - In "Step 1" on the page, go to "Google Cloud Platform Console".
- You will need a Google account to log in.

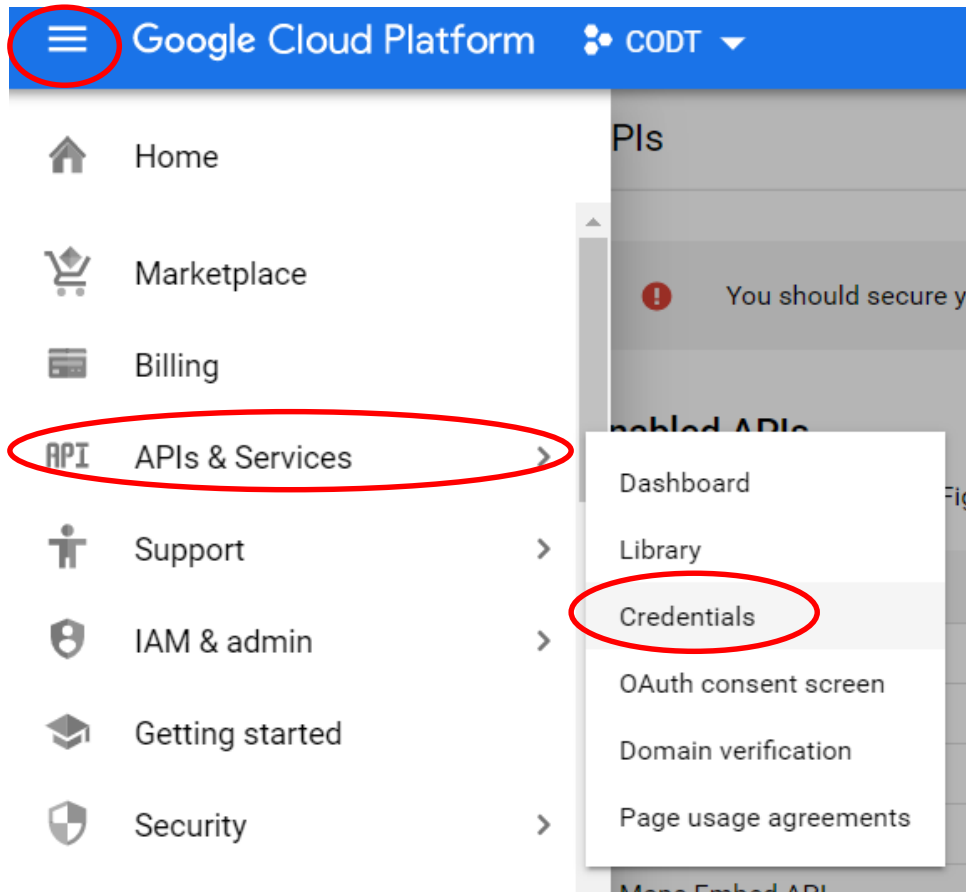
Get the API key

You must have at least one API key associated with your project.

To get an API key:

1. Go to the [Google Cloud Platform Console](#).
2. Click the project drop-down and select or create the project for which you want to add an API key.
3. Click the menu button  and select **APIs & Services > Credentials**.
4. On the **Credentials** page, click **Create credentials > API key**.
The **API key created** dialog displays your newly created API key.
5. Click **Close**.
The new API key is listed on the **Credentials** page under **API keys**.
(Remember to [restrict the API key](#) before using it in production.)

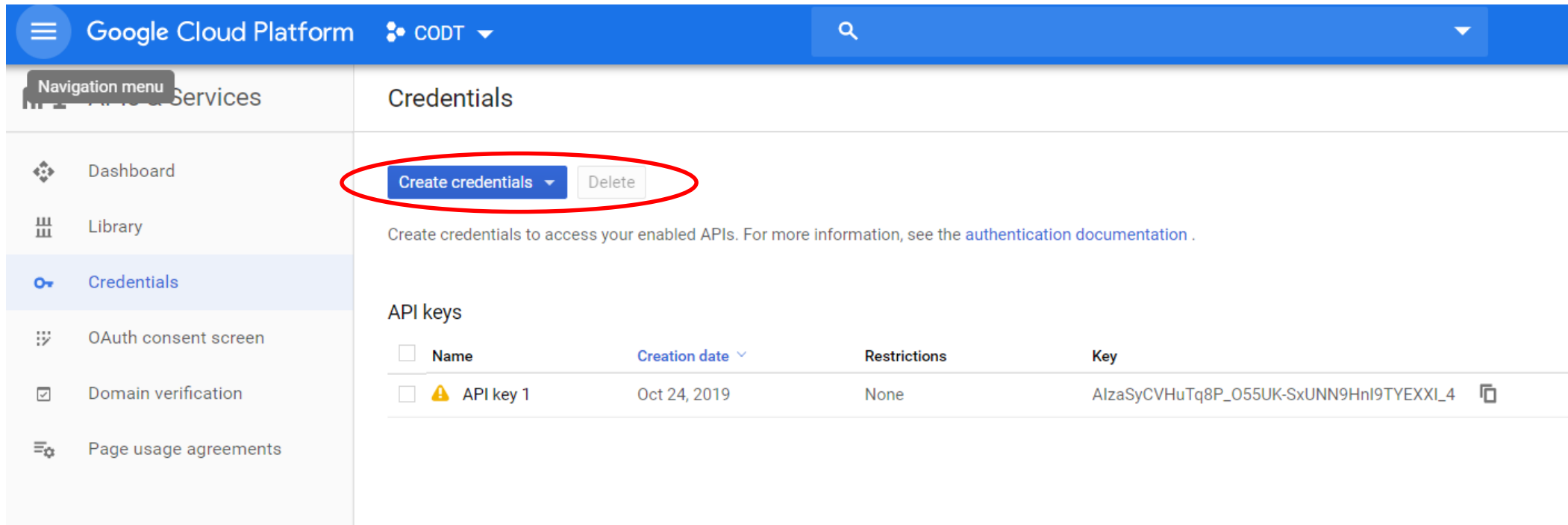
Obtaining API Key



Select "Credentials" to obtain an API.

* You need to create a project in Google Cloud Platform first

Obtaining API Key



The screenshot shows the Google Cloud Platform interface. The top navigation bar includes the Google Cloud Platform logo, a user profile icon labeled 'CODT', and a search bar. The left sidebar contains a 'Navigation menu' with options: Dashboard, Library, Credentials (selected), OAuth consent screen, Domain verification, and Page usage agreements. The main content area is titled 'Credentials'. It features a 'Create credentials' button (circled in red) and a 'Delete' button. Below these buttons, there is a text instruction: 'Create credentials to access your enabled APIs. For more information, see the [authentication documentation](#).' A table titled 'API keys' is displayed with the following columns: Name, Creation date, Restrictions, and Key. The table contains one entry: 'API key 1' with a creation date of 'Oct 24, 2019', no restrictions, and a key value 'AlzaSyCVHuTq8P_O55UK-SxUNN9Hni9TYEXXI_4'.

Name	Creation date	Restrictions	Key
API key 1	Oct 24, 2019	None	AlzaSyCVHuTq8P_O55UK-SxUNN9Hni9TYEXXI_4



Obtaining API Key

If you are unable to create an API, you may use the tutor's API key:

AlzaSyCVHuTq8P_O55UK-SxUNN9HnI9TYEXXI_4

Create a Simple Map

- Go to "Step 2" on the "Get API Key" page.

Adding a Map with a Marker
Marker Clustering
Importing Data into Maps
Visualizing Data
Combining Data
Mapping with Firebase
Geolocation
Using MySQL and PHP with Maps
Saving User-Added Data with Maps

Concepts
Overview
Map Types
Map and Tile Coordinates
Localizing the Map
Versioning

Styling a Map
Start Styling your Map
Custom Legends
Style Reference

Interacting with the Map
Controls
Events

[GET A KEY](#)

Notes:

- **Tip:** During development and testing, you can register a project for testing purposes in the Google API Console and use a generic, unrestricted API key. When you are ready to move your app or website into production, register a separate project for production, create a browser-restricted API key, and add the key to your application.
- **Premium Plan customers:** For production-ready apps, you must use a browser-restricted API key that is set up in the Google Maps APIs Premium Plan project created for you when you purchased the Premium Plan. Alternatively, you can [use a client ID](#) in combination with URL registration (instead of an API key).
- For more information, see the detailed guides below for [Standard Plan](#) and [Premium Plan](#) customers.

Step 2: Add the API key to your application

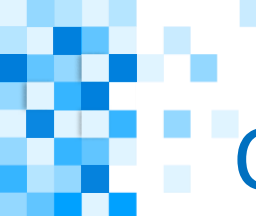
When loading the Google Maps JavaScript API, substitute `YOUR_API_KEY` in the code below with the API key you got from the previous step.

```
<script async defer src="https://maps.googleapis.com/maps/api/js?key=YOUR_API_KEY&callback=initMap" type="text/javascript"></script>
```

Copy this HTML line.

More about API keys

Detailed guide for users of the standard Google Maps JavaScript API
Detailed guide for users of the Google Maps APIs Premium Plan license
Choosing an authentication method for your application
Authenticating your application using an API key
Authenticating your application using a client ID
Specifying a client ID for the API key authorized



Create a Simple Map

contact.html

Open **contact.html** in your favourite editor and paste the copied line here.

Then replace the ***YOUR_API_KEY*** with the key that you obtained.

```
<script src="https://maps.googleapis.com/maps/api/js?key=AIzaSyCkMdZLiAUROgNlzsOsIItoZSRFer5MpM8&callback=initMap" async defer>
</script>

<script src="js/contact.js"></script>

</body>

</html>
```

You may use the tutor's API key: **AlzaSyCVHuTq8P_O55UK-SxUNN9HnI9TYEXXI_4**

Create a Simple Map

contact.html

Look for “my-map” id to make sure that this line is already added. If not please add here.

```
<div id="PageMain">
  <h1>We take great pride in our timely an
  email or over the phone.</h1>
  <h3>Two Trees Extra Virgin Olive Oil<br
  1750 Vine St<br />
  Los Angeles, CA</h3>
  <h3>info@twotreesoliveoil.com</h3>

  <div id="my-map"></div>

  <h2>Customer Service</h2>
  <div id="breadImage">
  <!-- a:link {
    color: #a8c686;
    text-decoration: none;
  }

  a:visited {
    text-decoration: none;
    color: #FFF;
  }

  a:hover {
    text-decoration: underline;
    color: #FC6;
  }

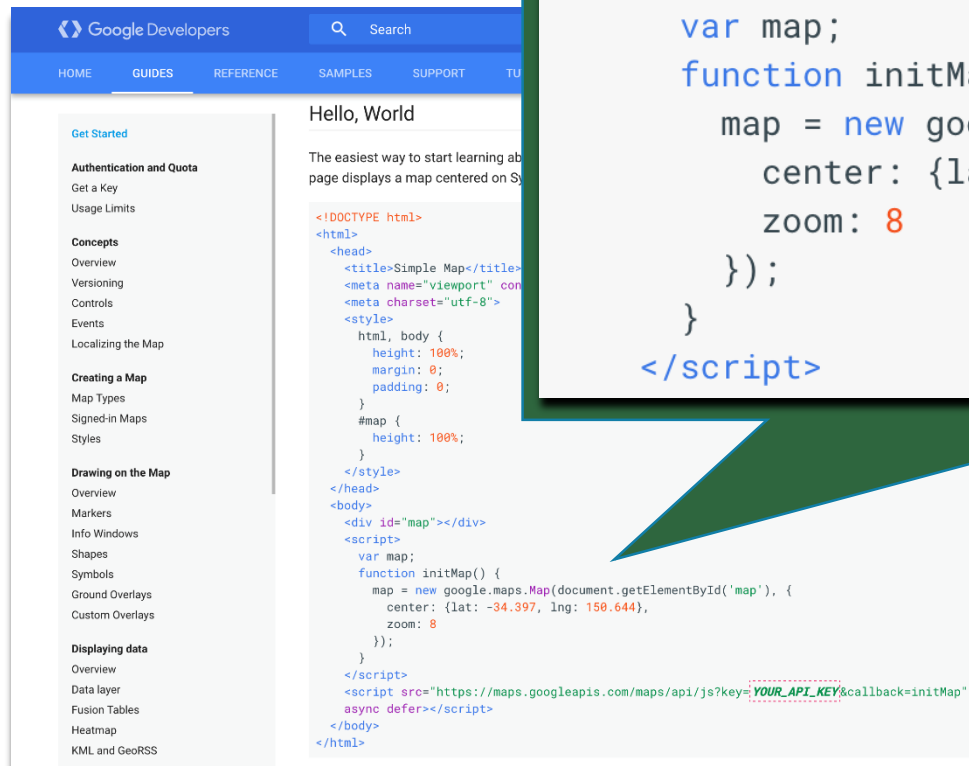
  a:active {
    text-decoration: none;
    color: #FC6;
  }

  #my-map {
    width: 500px;
    height: 400px;
    margin: 1em auto;
  }

-->
</style>
</head>
```

Create a Simple Map

Go back to the "Hello World" example in Google Maps developer page.



```
<script>
  var map;
  function initMap() {
    map = new google.maps.Map(document.getElementById('map'), {
      center: {lat: -34.397, lng: 150.644},
      zoom: 8
    });
  }
</script>
```

Copy this block of codes.

Create a Simple Map

contact.html

Paste the code here in the `contact.html` file.

```
Los Angeles, CA</h3>
<h3>info@twotreesoliveoil.com</h3>

<div id="my-map"></div>

<script>
  var map;
  function initMap() {
    map = new google.maps.Map(document.getElementById('my-map'), {
      center: {lat: -34.397, lng: 150.644},
      zoom: 8
    });
  }
</script>

<h2>Customer Service</h2>
<div id="breadImage">
<ul>
```

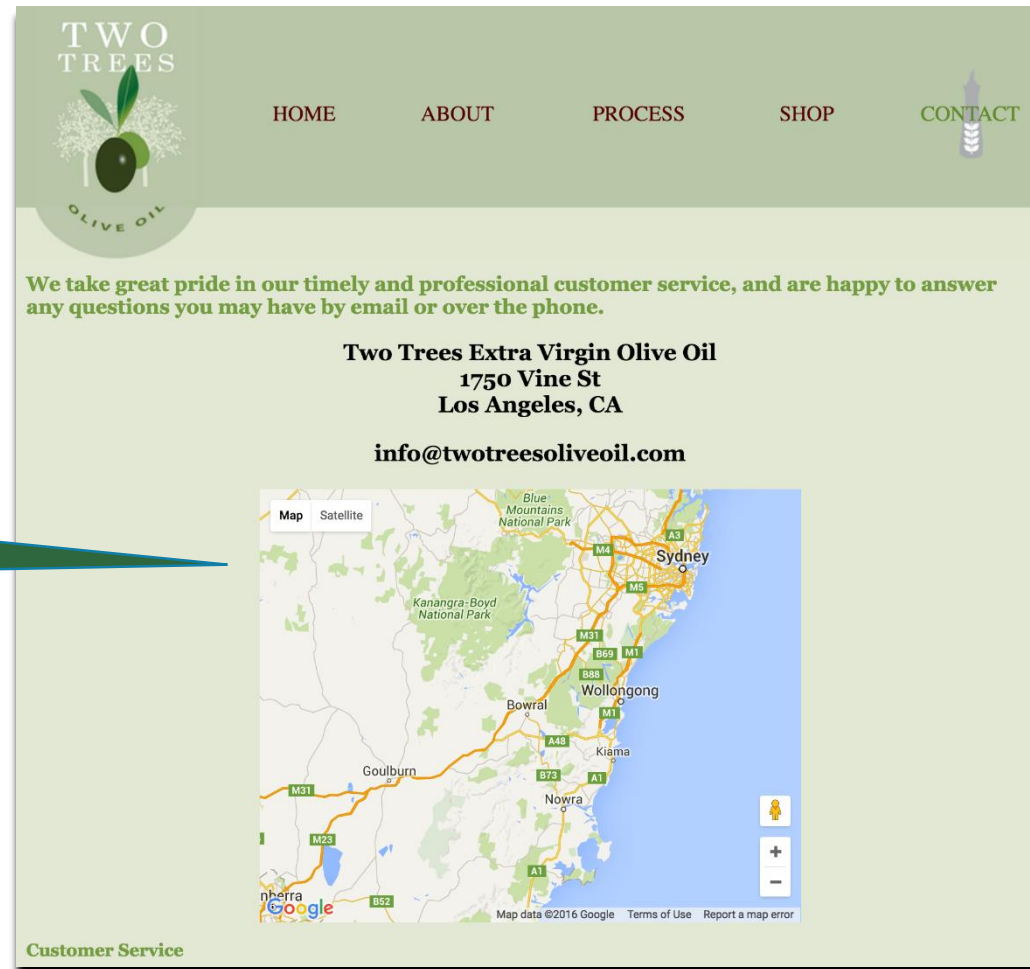
Change the original `map` to `my-map`.

Create a Simple Map

Open your browser and load the **contact.html** file.

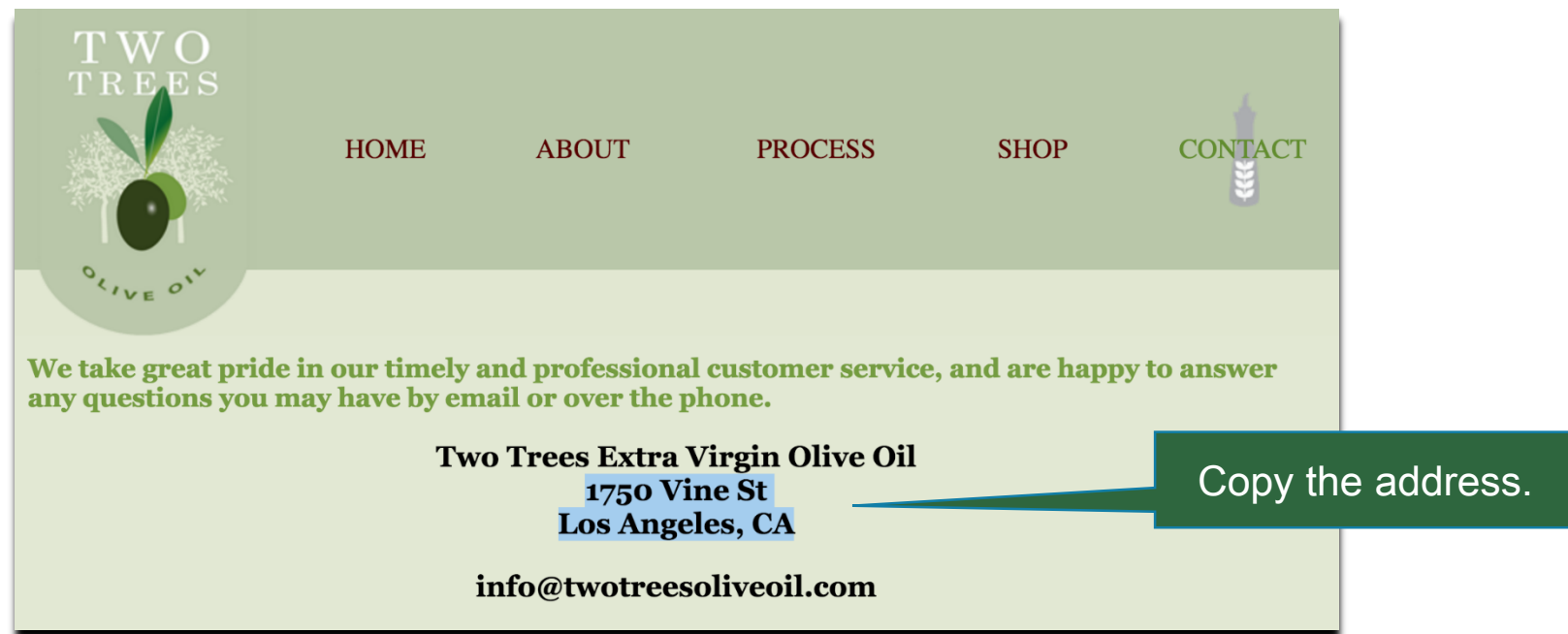
If everything is correct, a map will be loaded.

Next you will adjust the map to center at the address of the shop.



Changing the Map's Center Point

- To locate the latitude and longitude of a location, you can go to the GPS Visualizer web page:
 - <http://www.gpsvisualizer.com/geocode>



Changing the Map's Center Point

Convert an address to Latitude and Longitude at <https://www.latlong.net/convert-address-to-lat-long.html>

Get Lat Long from Address

By using this geographic tool you can get the **lat long coordinates from an a** of the city/town, state and street name to get more accurate lat long value. Als

Paste the address you copied previously in this textbox and click the "Geocode it" button.

Address

21 Tampines Avenue 1 Singapore 529757

Find

Write city name with country code for better results.

Latitude

1.346800

Longitude

103.931070

The result will show this. Copy the numbers here which are the latitude and longitude.

Changing the Map's Center Point

contact.html

```
Los Angeles, CA~/10/
<h3>info@twotreesoliveoil.com</h3>

<div id="my-map"></div>

<script>
  var map;
  function initMap() {
    map = new google.maps.Map(document.getElementById('my-map'), {
      center: {lat: 34.1031131, lng: -118.326343},
      zoom: 8
    });
  }
</script>
```

Replace the original latitude and longitude with the one that you copied.

Changing the Map's Center Point

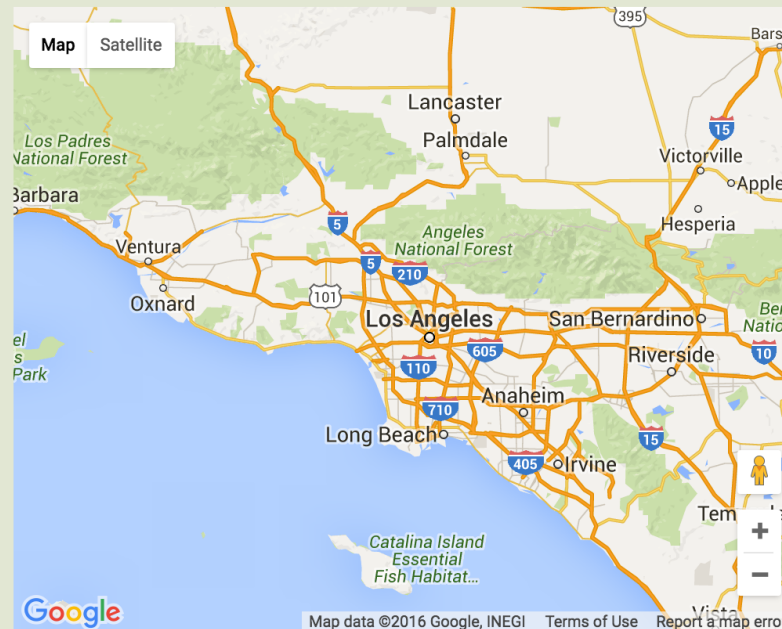
Open your browser and load the [contact.html](#) file.

The map now shows the correct location which is Los Angeles.

We take great pride in our timely and professional customer service, and are happy to answer any questions you may have by email or over the phone.

Two Trees Extra Virgin Olive Oil
1750 Vine St
Los Angeles, CA

info@twotreesoliveoil.com



Changing the Map Type and Zoom

contact.html

```
<h3>info@twotreesoliveoil.com</h3>

<div id="my-map"></div>

<script>
  var map;
  function initMap() {
    map = new google.maps.Map(document.getElementById('my-map'), {
      center: {lat: 34.1031131, lng: -118.326343},
      zoom: 14
    });
  }
</script>

<h2>Customer Service</h2>
```



Changing the Map Type and Zoom

Open your browser and load the [contact.html](#) file.

Notice the map is now zoomed in.



Adding a Marker to the Map

contact.html

Add this line.

Add this line.

Modify this line.

Add these lines.

```
<h3>info@twotreesoliveoil.com</h3>

<div id="my-map"></div>

<script>
  var map;
  var marker;
  function initMap() {

    var storeLocation = new google.maps.LatLng(34.1031131, -118.326343);

    map = new google.maps.Map(document.getElementById('my-map'), {
      center: storeLocation,
      zoom: 14,
      mapTypeId: google.maps.MapTypeId.ROADMAP
    });

    marker = new google.maps.Marker({
      position: storeLocation,
      map: map,
      title: "Find us here (Capital Records Tower for real)"
    });
  }
</script>

<h2>Customer Service</h2>
<div id="breadImage">info@twotreesoliveoil.com</h3>

<div id="my-map"></div>

<script>
  var map;
  var marker;
  var infoWindow;

  function initMap() {

    var storeLocation = new google.maps.LatLng(34.1031131, -118.326343);
    var popupContent = "Two Trees Extra Virgin Olive Oil<br>";
    popupContent += "1750 Vine St.<br>";
    popupContent += "Los Angeles, CA";

    map = new google.maps.Map(document.getElementById('my-map'), {
      center: storeLocation,
      zoom: 14,
      mapTypeId: google.maps.MapTypeId.ROADMAP
    });

    marker = new google.maps.Marker({
      position: storeLocation,
      map: map,
      title: "Find us here (Capital Records Tower for real)"
    });

    infoWindow = new google.maps.InfoWindow({
      content: popupContent,
      maxWidth: 270
    });

    google.maps.event.addListener(marker, 'click', function() {
      infoWindow.open(map, marker);
    });
  }
</script>

<h2>Customer Service</h2>
<div id="breadImage"> Maps JavaScript API' and buttons for 'GET A KEY' and 'VIEW'. A secondary navigation bar has tabs for 'OVERVIEW', 'GUIDES', 'REFERENCE' (which is highlighted), 'SAMPLES', and 'SUPPORT'. On the left, a sidebar lists 'API Reference - 3.exp (3.30)', 'API Reference - Release (3.30)', and 'API Reference - Frozen'. The main content area is titled 'Google Maps JavaScript API V3 Reference' with a five-star rating. Below the title is the text 'Experimental Version' and 'Last updated Tuesday, August 15, 2017'. A light blue box contains a star icon and a welcome message: 'Welcome to the reference for the experimental release of the Google JavaScript Maps API V3. This reference is kept up to date with the latest changes to the API. Please consult the [Javascript Maps API release notes](#) for information on what's new in each release.' The first section is 'Map class', which shows 'google.maps.Map' as a class that extends 'MVCObject'.

To know more about what Google Maps API can do, click on the "Reference" tab at the Google Maps developer web page.



CONGRATULATIONS!

# The End